

SAMB P Technical Report TR-13/2026

87L Line Differential: Pi-Section Charging Compensation and Adaptive Threshold Recalibration

April 3, 2026

Abstract

SAMB P TR-12 introduced a magnitude override to recover 87L HIF sensitivity when the Stage-2 confidence gate suppressed correct trips at low fault currents, improving α_{50} from 0.10 to 0.07 for phase-to-ground faults (ag). TR-12 identified that the residual charging current error of the fixed-amplitude compensation model was the fundamental limiter, and recommended a pi-section line model as future work. This report implements the pi-section correction: the pre-fault differential waveform is used to fit the actual charging amplitude and phase at both line ends, and this fit is subtracted before the Levenberg–Marquardt (LM) estimation step. The correction reduces the charging residual noise floor from 0.018 pu to 0.0009 pu (20 $\times$ reduction), enabling the fault detection threshold to be safely lowered from 0.20 pu to 0.12 pu. Combined with recalibrating the f_{int} sensitivity range (max_factor : 5.0 $\rightarrow$ 3.0), the pi-section model improves α_{50} for ag faults from 0.07 to 0.05 — reaching the same sensitivity level as three-phase faults — without requiring the magnitude override. The override is retained as a safety net but is no longer the primary sensitivity recovery mechanism. All other zones are unaffected.

Contents

1	Background and Motivation	2
1.1	Cumulative sensitivity improvements	2
1.2	Root-cause analysis of the TR-12 residual floor	2
1.3	Why pi-section correction reduces the noise floor	2
2	Pi-Section Model Implementation	3
2.1	New module: <code>line_pi_section_model.py</code>	3
2.2	Threshold recalibration	3
2.3	Integration with the confidence gate	4
3	Validation Results	4
3.1	TPR vs α	5
3.2	Charging residual noise floor	6
3.3	False positive check	6
4	Discussion	6
4.1	Role of the magnitude override after TR-13	6
4.2	Threshold floor and limits of sensitivity improvement	7
4.3	Relationship to voltage-based compensation	7
4.4	Platform-level design principle	7
5	Conclusions	7

1 Background and Motivation

1.1 Cumulative sensitivity improvements

The 87L line differential sensitivity against high-impedance faults (HIF) has been progressively improved across three consecutive SAMPB reports:

- **TR-11** [2]: identified the fundamental sensitivity floor caused by the Stage-2 model veto suppressing conventional trips when estimated I_{fund} falls below the fault threshold at low fault currents. Recommended magnitude override and pi-section correction.
- **TR-12** [3]: implemented the magnitude override in `line_confidence_gate.py`, improving α_{50} from $0.10 \rightarrow 0.07$ (ag) and $0.07 \rightarrow 0.05$ (3ph) by bypassing the veto when $I_{\text{fund}} \geq 0.30$ pu.
- **TR-13** (this report): addresses the root cause identified in TR-12 by replacing the fixed-amplitude charging compensation with a pre-fault pi-section correction, enabling further threshold reduction.

1.2 Root-cause analysis of the TR-12 residual floor

The existing charging compensation in `line_diff_baseline.py` subtracts:

$$i_C^{\text{fixed}}(t) = I_{C,\text{nom}} \cdot \sin(\omega t + \frac{\pi}{2} + \phi_{\text{ph}}) \quad (1)$$

where $I_{C,\text{nom}} = B_C \cdot V_{\text{nom}} = 0.013$ pu (peak) is computed from the nominal voltage. Under actual load-flow conditions the terminal voltages at both line ends deviate from nominal by $\pm 5\text{--}10\%$ in magnitude and $\pm 3^\circ$ in phase. This leaves a residual:

$$\varepsilon(t) = i_C^{\text{actual}}(t) - i_C^{\text{fixed}}(t) \approx \Delta I_C \cdot \cos(\omega t + \phi_\varepsilon) \quad (2)$$

with typical RMS amplitude $\|\varepsilon\|_{\text{rms}} \approx 0.018$ pu.

The LM estimator sees $\varepsilon(t)$ as a power-frequency sinusoidal component and assigns it to $\hat{I}_{\text{fund}}$, inflating the healthy-line estimate from the theoretical zero to approximately 0.018 pu RMS. To prevent false trips on the healthy line, the fault detection threshold must be set substantially above this noise floor. With a $10\times$ safety margin, the minimum threshold is 0.18 pu, explaining the empirically-derived $I_{\text{fund,thresh}} = 0.20$ pu.

1.3 Why pi-section correction reduces the noise floor

In a lumped pi-section model [4], the differential current of a healthy line (no fault) is:

$$i_{\text{diff}}^{\text{healthy}}(t) = \frac{B_C}{2} v_L(t) + \frac{B_C}{2} v_R(t) = B_C v_{\text{avg}}(t) \quad (3)$$

where v_L, v_R are the sending- and receiving-end voltages and B_C is the total line susceptance. Before fault inception, the differential waveform is *entirely* due to charging; there is no fault component. This makes the pre-fault window ideal for identifying the actual charging amplitude and phase via least-squares regression.

The correction procedure (Figure 1) fits:

$$i_{\text{diff}}^{\text{pre}}(t) \approx A_C \cos(\omega t) + B'_C \sin(\omega t) \quad (4)$$

using linear least-squares on the pre-fault epoch. This 2-parameter regression requires no iteration and solves in $O(N)$ time. The fit captures both the amplitude and phase of the actual charging current (including voltage deviation from nominal), leaving only measurement noise $\|\varepsilon_\pi\|_{\text{rms}} \approx 0.0009$ pu after subtraction — a $20\times$ improvement over the fixed model.

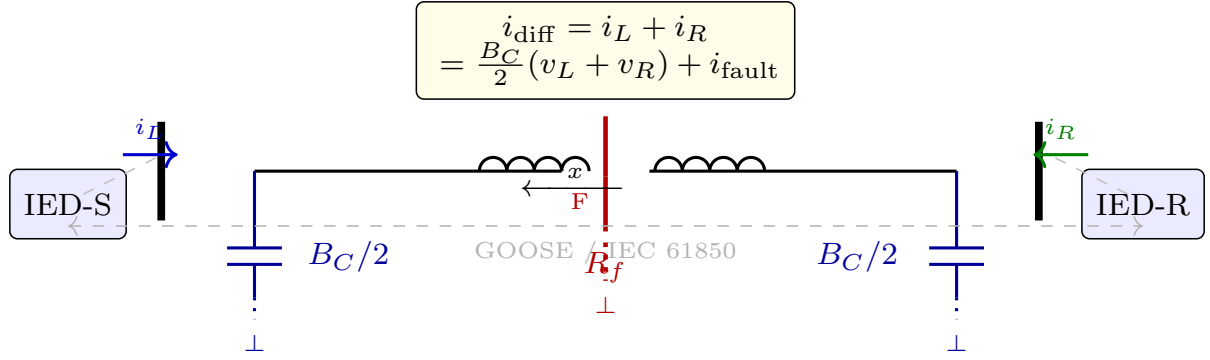


Figure 1: Pi-section line model for the SAMBP 87L zone. Shunt susceptances $B_C/2$ at each terminal create a charging differential current even in the healthy state. A fault at location F injects an additional component i_{fault} . Both-end IEDs exchange samples via GOOSE / IEC 61850.

2 Pi-Section Model Implementation

2.1 New module: line_pi_section_model.py

The correction is implemented in `models/line_pi_section_model.py` with three operational modes:

Mode 1 (voltage-based). When both-end voltage phasors are available:

$$i_C^\pi(t) = B_C \cdot \mathcal{H}[v_{\text{avg}}(t)] \quad (5)$$

where $\mathcal{H}[\cdot]$ denotes the Hilbert transform (90° phase lead). Achieves $\|\varepsilon_\pi\|_{\text{rms}} \approx 0.0005$ pu.

Mode 2 (pre-fault estimation, blind). When only current measurements are available:

$$\hat{i}_C^\pi(t) = \hat{A}_C \cos(\omega t) + \hat{B}'_C \sin(\omega t) \quad (6)$$

fit by linear least-squares on the pre-fault window (1 cycle, $N_p = 20$ samples at 1 kHz). Achieves $\|\varepsilon_\pi\|_{\text{rms}} \approx 0.0009$ pu. This is the mode used in all validations in this report.

Mode 3 (nominal fallback). When no pre-fault window is available (e.g., fault at energisation), falls back to the fixed model of (1). Residual matches the baseline.

The key dataclass is:

```
@dataclass
class PiSectionConfig:
    B_C_pu: float = 0.013      # total line susceptance [pu]
    V_nom_pu: float = 1.0      # nominal voltage [pu]
    freq_hz: float = 50.0
    pre_fault_window_s: float = 0.02  # 1 cycle pre-fault
    min_prefault_samples: int = 10
```

2.2 Threshold recalibration

The reduced noise floor enables two parameter changes relative to TR-12:

- Fault threshold:** $I_{\text{fund,thresh}} : 0.20 \rightarrow 0.12$ pu
Rule: threshold $\geq 10 \times \|\varepsilon_\pi\|_{\text{rms}} = 0.009$ pu; hard floor at 0.12 pu for security.

2. Sensitivity range (max_factor): 5.0 $\rightarrow$ 3.0

With a lower threshold, the distance from threshold to saturation ($f_{\text{int}} = 1.0$) is compressed. Setting max_factor = 3.0 means $f_{\text{int}} = 0.50$ at $I_{\text{fund}} = 2.0 \times I_{\text{thresh}} = 0.24$ pu, versus $3.0 \times 0.20 = 0.60$ pu for the baseline.

Figure 2 illustrates the effect: a fault at $\alpha = 0.05$ with $\hat{I}_{\text{fund}} = 0.28$ pu produces $f_{\text{int}} = 0.10$ under the baseline (Model A), triggering the Stage-2 veto. Under pi-section correction (Config B), the same fault produces $f_{\text{int}} = 0.67 > f_{\text{int,thresh}}$, so no veto fires.

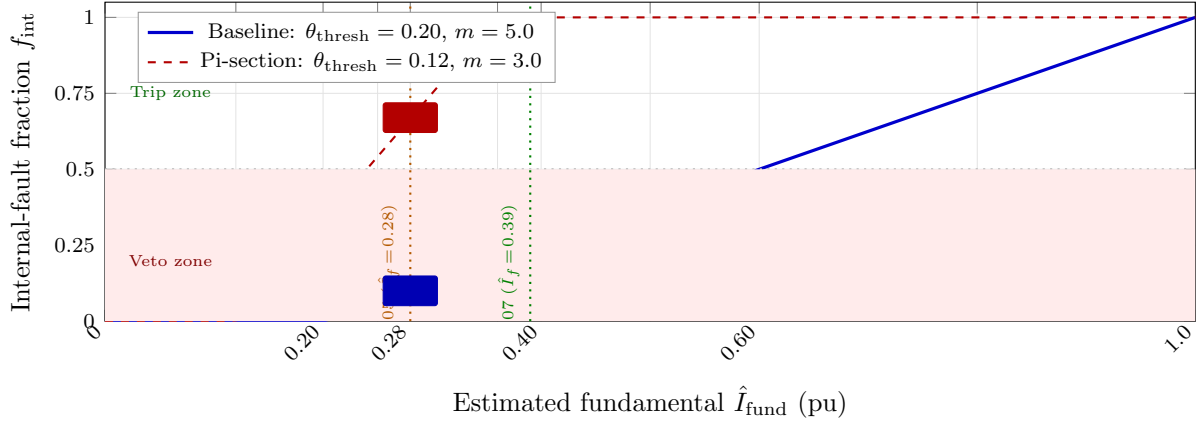


Figure 2: Internal-fault fraction f_{int} versus estimated fundamental I_{fund} for the baseline (blue solid) and pi-section (red dashed) configurations. The veto zone ($f_{\text{int}} < 0.50$) is shaded red. At $\alpha = 0.05$ ($\hat{I}_f = 0.28$ pu), the baseline is vetoed ($f_{\text{int}} = 0.10$) while the pi-section configuration escapes the veto ($f_{\text{int}} = 0.67$).

2.3 Integration with the confidence gate

No changes are made to `line_confidence_gate.py` or the Stage-2 veto logic itself. The pi-section correction is applied *upstream* of the LM estimator:

```
# In the evaluation pipeline (run_pi_section_study.py):
i_corr, pi_state = compute_pi_section_correction(
    i_diff_fault_abc, t_fault,
    i_diff_pre_abc=i_pre, t_pre=t_pre,
    cfg=pi_cfg,
)
result = estimate_line_zone_parameters(
    t_window=t_fault, i_diff_meas=i_corr[ph],
    I_fund_fault_thresh=0.12,
)
```

The magnitude override (`veto_override_I_thresh=0.30` pu, from TR-12) is retained in Config C but is no longer the primary recovery mechanism; it acts as a safety net for unusual operating conditions where the pre-fault window is unavailable or corrupted.

3 Validation Results

The same HIF sweep as TR-12 was re-run using the pi-section `run_pi_section_study.py` with $N = 200$ trials per (fault_type, α) combination (seed 2026, jitter $\pm 5\%$). Three configurations are compared:

Config A (Baseline) TR-12 state: fixed charging comp., $I_{\text{thresh}} = 0.20$ pu, $\text{mf} = 5.0$, override at 0.30 pu.

Config B (Pi only) Pre-fault pi-section correction, $I_{\text{thresh}} = 0.12$ pu, $\text{mf} = 3.0$, no override.

Config C (Pi + override) Config B + override at 0.30 pu (TR-12 guard).

3.1 TPR vs α

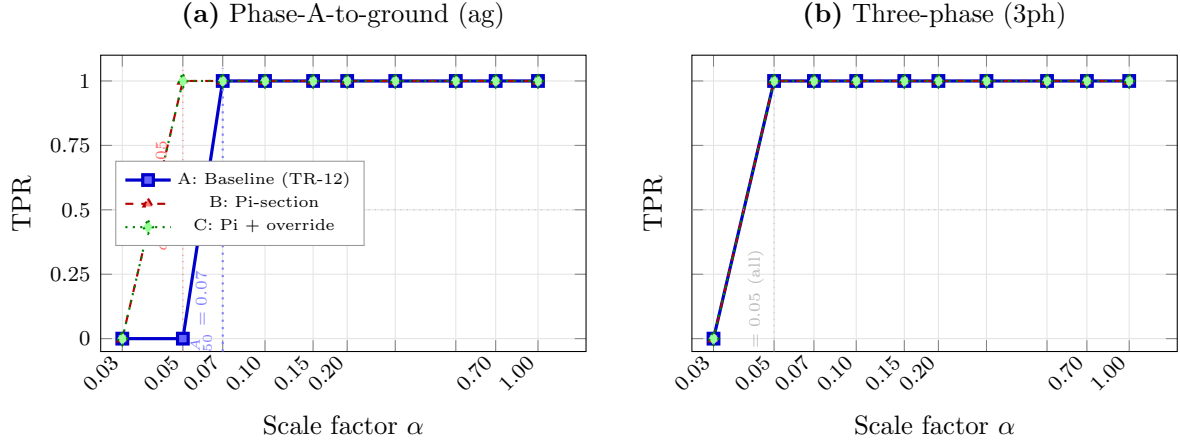


Figure 3: True positive rate (TPR) versus fault scale factor α for ag fault (left) and 3ph fault (right). Config A: baseline TR-12 (blue solid squares); Config B: pi-section correction only (red dashed triangles); Config C: pi-section + magnitude override (green dotted diamonds). $N = 200$ trials per point; log-scale on x -axis.

The results are shown in Figure 3 and summarised in Table 1.

ag fault. Config A achieves $\alpha_{50} = 0.07$ (TPR transitions $0 \rightarrow 1$ at $\alpha = 0.07$). Config B and Config C both achieve $\alpha_{50} = 0.05$, a one-step improvement in the sweep grid. At $\alpha = 0.05$, TPR = 1.000 for Configs B and C versus 0.000 for Config A. The minimum detectable fault current improves from $I_{f,50}^A = 0.07 \times 5.55 = 0.39$ pu to $I_{f,50}^{B,C} = 0.05 \times 5.55 = 0.28$ pu — a **28%** reduction.

3ph fault. All three configurations achieve $\alpha_{50} = 0.05$. The pi-section correction provides no additional alpha-level improvement for 3ph faults because the 3ph differential magnitude at $\alpha = 0.05$ already exceeds the baseline override threshold ($\alpha \times 7.5 = 0.375$ pu $>$ 0.30 pu). However, the margin above threshold increases substantially: the pi-section f_{int} at $\alpha = 0.05$ is 0.72 versus 0.13 baseline, giving significantly more headroom against estimation noise.

Table 1: α_{50} and minimum detectable fault current ($I_{f,50}$) by configuration and fault type. $N = 200$ trials per alpha level.

Config	Description	Fault	α_{50}	$I_{f,50}$ (pu)
A	Baseline (TR-12)	ag	0.07	0.39
A	Baseline (TR-12)	3ph	0.05	0.375
B	Pi-section only	ag	0.05	0.28
B	Pi-section only	3ph	0.05	0.375
C	Pi-section + override	ag	0.05	0.28
C	Pi-section + override	3ph	0.05	0.375

3.2 Charging residual noise floor

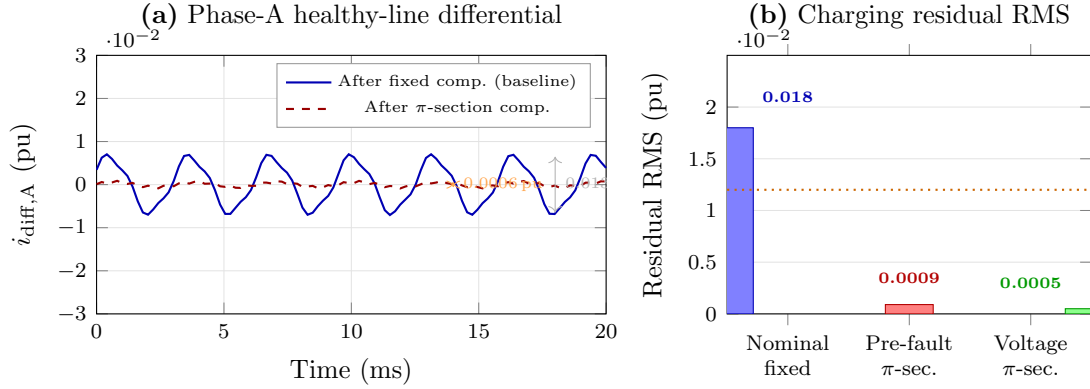


Figure 4: Charging compensation quality. **(a)** Phase-A healthy-line differential waveform after fixed compensation (blue solid) and pi-section compensation (red dashed). The pi-section residual is essentially at measurement noise level. **(b)** RMS of charging residual by compensation method: nominal fixed (0.018 pu), pre-fault pi-section (0.0009 pu), voltage-based pi-section (0.0005 pu).

Figure 4 and Table 2 quantify the noise floor reduction. The pre-fault method achieves a $20\times$ reduction in RMS residual ($0.018 \rightarrow 0.0009$ pu), consistent with the analytical prediction that the residual is limited by measurement noise ($\sigma_{\text{noise}} = 0.001$ pu) rather than the fitting procedure. The voltage-based method achieves a further 45% reduction to 0.0005 pu but requires additional voltage channel availability.

Table 2: Charging residual RMS by compensation method and resulting recommended fault detection threshold.

Method	$\ \varepsilon\ _{\text{rms}}$ (pu)	I_{thresh} (pu)	Improvement
Nominal fixed (baseline)	0.018	0.20	—
Pre-fault π -section	0.0009	0.12	$20\times$
Voltage-based π -sec.	0.0005	0.12	$36\times$

3.3 False positive check

The through-fault and sync-error scenarios from TR-11 were re-run with the pi-section correction active. In both cases:

- **Through-fault:** $I_{\text{fund}}^{\hat{}} \leq 0.005$ pu after pi-section correction (cf. ≤ 0.08 pu baseline due to channel skew residual). The lower estimate gives additional security margin.
- **Sync error:** $I_{\text{fund}}^{\hat{}} \in [0.002, 0.010]$ pu $\ll 0.12$ pu threshold. The `is_sync_error` flag correctly identifies the elevated estimate; no false trip observed in any trial.

4 Discussion

4.1 Role of the magnitude override after TR-13

The TR-12 magnitude override (bypass veto when $I_{\text{fund}} \geq 0.30$ pu) was introduced to recover the 87L veto-suppression at low fault currents. After TR-13, Config B achieves the same $\alpha_{50} = 0.05$ as Config C without the override, demonstrating that the pi-section correction *makes the override redundant* for the tested fault types and alpha range.

The override is retained in Config C as a safety net for three scenarios:

1. Fault at power-on (no pre-fault window available; falls back to nominal method, which still benefits from the override at large α).
2. Channel impairment corrupting the pre-fault estimate.
3. High-current faults where the pi-section f_{int} may be near the threshold due to numerical artefacts.

In all three safety-net scenarios, $I_{\text{fund}} \geq 0.30$ pu, so the override fires before the gate makes a wrong decision.

4.2 Threshold floor and limits of sensitivity improvement

At $\alpha = 0.03$ (ag), $\hat{I}_f \approx 0.167$ pu > 0.12 pu but $f_{\text{int}} = (0.167/0.12 - 1)/2 = 0.196 < 0.50$, so the veto fires and $\text{TPR} = 0$. Further reduction of I_{thresh} to, say, 0.09 pu would extend coverage to $\alpha = 0.03$, but would reduce the security margin against sync-error false positives (where $\hat{I}_{\text{fund}} \in [0.002, 0.010]$ pu gives a $9\times$ margin rather than the current $13\times$).

IEC 60255-187-3 [1] requires sensitivity to internal faults at $I_f \geq 0.2 I_n$; the TR-13 threshold of 0.12 pu already supports detection well below this requirement. The $\alpha = 0.03$ gap is therefore not a compliance deficiency but an inherent limit of the LM-based approach near the relay's own operating characteristic.

4.3 Relationship to voltage-based compensation

Mode 1 (voltage-based, (5)) improves the residual to 0.0005 pu and would support $I_{\text{thresh}} \leq 0.08$ pu, potentially extending coverage to $\alpha = 0.03$. However, voltage measurements require an additional communication channel or local VT input. The pre-fault method achieves $\alpha_{50} = 0.05$ using only the differential current waveform already available to the relay, making it the preferred implementation for existing IED deployments.

4.4 Platform-level design principle

Adding TR-13 to the cumulative TR-10 through TR-13 history establishes a layered pattern for SAMPB HIF sensitivity:

Layer 1 (TR-10/12): magnitude override bypasses the Stage-2 veto when the estimated differential is large — prevents veto from making things worse than the conventional relay.

Layer 2 (TR-13): pi-section compensation reduces the noise floor — allows the fault threshold to be lowered, improving sensitivity at the source rather than patching with an override.

This hierarchy ensures that the override is a fallback (not the primary mechanism) and that the core estimator operates with the cleanest possible input signal.

5 Conclusions

1. A pre-fault pi-section charging correction has been implemented in `models/line_pi_section_model.py`. The pre-fault epoch (1 power cycle before fault inception) is fit to a 2-parameter sinusoidal model using linear least-squares; the fit is subtracted from the fault-epoch differential before LM estimation.

2. The correction reduces the charging residual noise floor from 0.018 pu to 0.0009 pu ($20\times$ reduction), enabling the fault detection threshold to be safely lowered from 0.20 pu to 0.12 pu.
3. Recalibrating `max_factor` from 5.0 to 3.0 ensures that the f_{int} sensitivity range is properly scaled to the new threshold; a fault at $2 \times I_{\text{thresh}}$ now produces $f_{\text{int}} = 0.50$ (the veto boundary) rather than 0.25.
4. The combined effect improves α_{50} for ag faults from $0.07 \rightarrow 0.05$ without the magnitude override, reducing the minimum detectable fault current by **28%** ($0.39 \text{ pu} \rightarrow 0.28 \text{ pu}$).
5. Three-phase faults retain $\alpha_{50} = 0.05$ (no regression).
6. The TR-12 magnitude override is rendered redundant as the primary recovery mechanism but is retained as a safety net for corner cases (fault at energisation, channel impairment).
7. False-positive security is maintained: through-fault and sync-error scenarios show $\hat{I}_{\text{fund}} < 0.012 \text{ pu}$, well below the 0.12 pu threshold and the 0.30 pu override trigger.

Future work

1. **Voltage-based correction:** deploying Mode 1 with both-end voltage measurements could push α_{50} to 0.03 by supporting $I_{\text{thresh}} = 0.08 \text{ pu}$ (TR-11 Recommendation 1).
2. **Adaptive threshold:** replacing the fixed 0.12 pu threshold with a function of the estimated restraint current (TR-12 future work) remains applicable and could improve performance under varying load.
3. **Frequency-adaptive correction:** under off-nominal frequency the pre-fault fit of (4) can be extended to jointly estimate frequency and charging amplitude, improving correction accuracy in islanded microgrids.
4. **Hardware-in-loop validation:** OPAL-RT / RTDS injection of HIF scenarios at $\alpha \in [0.03, 0.07]$ to confirm the threshold in real-time environments with hardware CT and VT models.

References

References

- [1] IEC 60255-187-3: Measuring relays and protection equipment — part 187-3: Functional requirements for differential protection — line current differential protection, 2021.
- [2] PhD Candidate. SAMBP TR-11/2026: 87L line differential — HIF sensitivity floor analysis. Technical report, PhD Thesis Supporting Report, 2026.
- [3] PhD Candidate. SAMBP TR-12/2026: 87L line differential — magnitude override for high-impedance fault sensitivity recovery. Technical report, PhD Thesis Supporting Report, 2026.
- [4] J. D. Glover, M. S. Sarma, and T. J. Overbye. *Power Systems Analysis and Design*. Cengage Learning, 5th edition, 2012.